Myong Jong Shin

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Profile

Economics Ph.D. specializing in time-series econometrics and the evaluation of macroeconomic and financial forecasts. Macroeconomic research focuses on forecasting US industrial production growth and inflation. Currently a Quantitative Modeling Sr. Associate at JPMorgan Chase, building and validating statistical and machine learning models on large-scale data and writing validation reports for regulators, including the OCC. Experienced in explaining technical results to non-specialist and expert audiences alike.

Education

Indiana University Bloomington (Graduate Assistantship; H. Crawford Graduate Top-up Fellowship)
Ph.D. in Economics Sept 2016 – Dec 2023
Dissertation: Three Essays on Model Comparisons in Finance and Macroeconomics
M.A. in Economics Sept 2016 – Mar 2019

Soongsil University (Soongsil Scholarship)
B.A. in Economics Mar 2010 – Dec 2015

Experience

JPMorgan Chase, New York May 2023 – Present
Quantitative Modeling Sr. Associate

- Develop benchmark machine learning models (tree-based models and LLMs) for anti-money laundering applications, including transaction-alert analysis, sanctions screening and surveillance, in line with federal AML regulations.
- Designed a statistical framework that determines the sample sizes needed for statistical inference on high-volume transaction data; the framework is used across transaction monitoring and sanctions screening.
- Author model validation reports for review by US banking regulators, including the OCC, translating technical statistical findings into clear written conclusions.
- Assess ML models in production for statistical soundness and interpretability.

Taylor & Francis Jan 2024 – Present
Journal Reviewer, Econometric Reviews

- Review manuscripts submitted to Econometric Reviews.

Indiana University Bloomington Sept 2018 – May 2022
Associate Instructor and Teaching Assistant

- Taught undergraduate business statistics, micro- and macroeconomics, and graduate financial econometrics; delivered lectures and designed exams and assignments.
- Led recitations for Ph.D. econometrics (bootstrap, instrumental variables, causal effects).

Research

Test of Equal Predictive Ability with HAR Standard Errors for Forecasting US Industrial Production Growth and Inflation (working paper)

- Forecast US industrial production growth and inflation with autoregressive distributed lag (ADL) models built on predictors from the FRED-MD monthly macroeconomic database, benchmarked against an autoregressive model.
- Applied heteroskedasticity- and autocorrelation-robust (HAR) standard errors to tests of equal predictive ability, improving size and power in small samples.
- **Finding:** HAR-based tests are more conservative, with fewer ADL models showing better predictive ability than the autoregressive benchmark.

Empirical Investigation of Supervised Machine Learning Models Predicting the Equity Risk Premium (job market paper; circulating for journal submission)

- Forecast firm-level US equity risk premia at monthly to annual horizons (1987–2019) with principal component regression, partial least squares, LASSO, ridge, elastic net and random forests, using CRSP data starting from 1960.
- Evaluated forecasts with the Superior Predictive Ability test, the Model Confidence Set, and a conditional SPA test that conditions on indicators of the state of the US macroeconomy and financial markets.
- **Finding:** No model dominates across sectors, and the best-performing model differs by industry (e.g., random forests fit finance, insurance and real estate well but not public administration). Weekly and monthly price trends (price momentum) contribute most to the random forest's predictions.

Evaluation of Supervised Machine Learning Models Predicting the Equity Risk Premium in South Korea (circulating for journal submission)

- Applied a similar forecasting and evaluation framework from the job market paper to Korean equities, using Worldscope data on a broad firm universe to avoid survivorship bias.
- **Finding:** No model dominates, and weekly and monthly price trends contribute most to random forest predictions.

Selected Project

JPMC Systematic Trading Competition 2025

- Ranked 8th of 92 participants in October; built a portfolio by forecasting stock returns from price signals decomposed by Fourier frequency and selected with L2 (ridge) penalization.

Skills

- **Macroeconomics & Forecasting:** US macroeconomic forecasting, monetary economics, mixed-frequency models (MIDAS), dynamic factor models, DSGE modeling, structural estimation
- **Econometrics & Statistics:** time-series econometrics, forecast evaluation (equal and superior predictive ability tests, Model Confidence Set), out-of-sample testing, hypothesis testing, power analysis, stochastic calculus
- **Machine Learning:** ensemble methods, gradient boosting, neural networks, regularization, LLMs for sentiment analysis
- **Programming:** Python (PyTorch, TensorFlow), R, Stata, MATLAB, C/C++, Bash, LaTeX, Git
- **Data:** FRED-MD and FRED-QD, Bloomberg, Datastream, WRDS (CRSP), Worldscope